06th September 2021 Crop science Test 1

Section A

1.The benefit of Carbon 3 plants in countries with high carbon dioxide emission is due to…

A. high saturation point **B**. no calvin cycle **C**. low saturation point **D**. low carbo intake

2.Which respiration path way is located in both the cytosol and plastid

A. Glycolysis **B**. pentose phosphate **C**. carbon dioxide **D**. citric acid

3.Which respiration process occurs in the mitochondria of the plant cell

A. Glycolysis **B**. pentose phosphate **C**. carbon dioxide **D**. citric acid

4.The chemical compounds in the plant that inhibit growth of surrounding plants

A. Terpenes **B**. alkaloids **C**. Tannins **D**. nitric oxide

5.What makes the stem cell to be unique is that the do not

A. divide **B**. produce daughter cells C. differentiate D. grow

Section B

6. What properties of water makes it to be the best solvent to be used by plants **2 marks**

7. What is cavitation? How does the plant overcome the problem? **2 marks**

8. A vegetable producer insists to apply the foliar fertilizer on the plant, but why should you discourage that farmer to use foliar fertilizer? **1 mark**

9. Given the planting spacing of maize at 30cm (0.3m) by 75cm (0.75m), the farmer decides to apply 4.5g per plant in one hectare, how many 50kg bags can be recommended to buy? **2marks**

10. Describe the photosystems **3 marks**

11. Upon hearing that you have completed a crop science course, a neighbour comes to you to ask for help. He got leaf samples from the tomato plant, unfortunately he lost the labels of the sample and the only information he could provide is that one leaf was taken from the plant that was under shade, how could you help to identify the leaf the came from the area under shade. **2 marks**

12. What environmental factors can affect the process of respiration **2 marks**

13. Why do plants produce the secondary metabolites **1 mark**

14. Explain the main function of phytochrome to the plant environment **1 mark**

15. A fruit tree farmer buys the hormone auxins to use at the farm, how would you advise the farmer to use the hormone? **2 marks**

SECTION A

1. A ✓
2. B ✓
3. D ✓
4. A ✗ C
5. D ✗ C

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SECTION B

6. i. Water is a polar solvent, that can uptake sucrose via the phloem tissue
   ii. Water has cohesive properties. ~~that enables~~

7. Cavitation is the formation of bubbles in a fluid that disrupts the flow of water in plants. (1)

8. (i) Foliar fertilizers are relatively expensive than dry fertilizers.
   (ii) Provided that the plant already has enough NPK, application of foliar fertilizers will have no effect
   (iii) Folliar fertilizers may cause leaf scorching on plants if it is applied in excess. (1)

9. Plant geometry = Intra row × Inter row
   = 0.3m × 0.75m
   = 0.225 m^2

1000g — 1kg
4.5g — x

$$\frac{1000g(x)}{1000g} = \frac{4.5g \times 1kg}{1000g}$$

$$x = 4.5 \times 10^{-3} kg$$

1

4.5×10^{-3} kg — 0.225 m^2

x — 1 hectare (10 000 m^2)

4.5×10^{-3} kg / x — 0.225 m^2 / 10 000 m^2

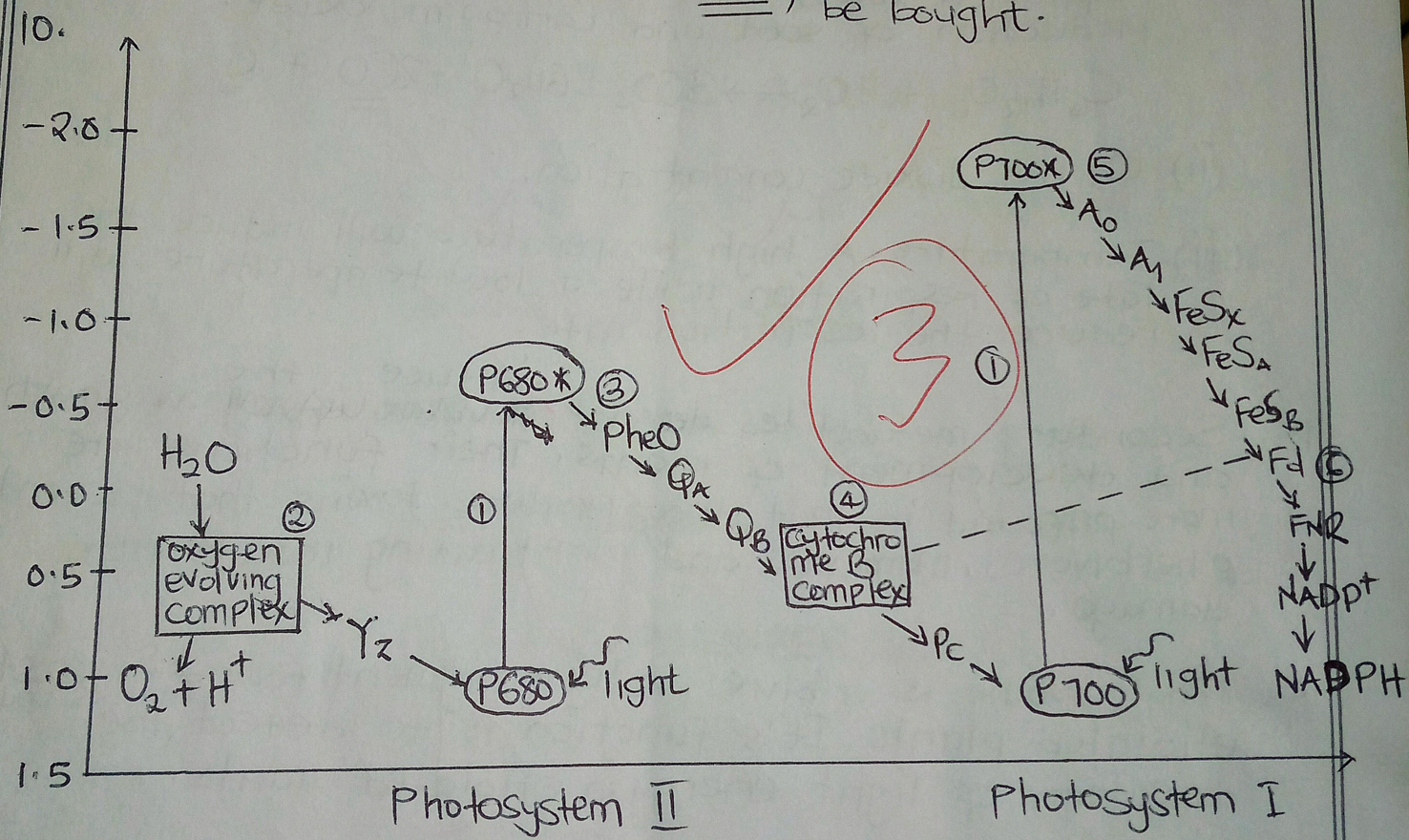
$\frac{0.225\,m^2}{0.225\,m^2}(x) = \frac{10000\,m^2 \times 4.5 \times 10^{-3}\,kg}{0.225\,m^2}$

$x = \frac{45\,kg}{0.225}$

$x = 200\,kg$

∴ Number of 50kg bags = $\frac{200kg}{50kg}$

= 4, will be recommended to be bought.



- With the aid of the diagrams above;
- Photosystem I involves the reduction of $NADP^+$ to NADPH
- The produced electron is a strong oxidant.
- A photon of light is absorbed at 700 nm.

- Photosystem II involves the oxidation of a water molecule to produce oxygen.
- A photon of light is absorbed at 680 nm.
- However, electrons play an important role in both the photosystems.

11. (i) The leaf will not exhibit the green colour as opposed to that which was not under shade.
(ii) The leaf will be folded as there will be insufficient or no light to be trapped by it.

12. (i) Environmental oxygen. When there is insufficient oxygen available for glucose breakdown, the overall respiration process will be affected by an additional production of soot and carbon monoxide.

$$C_6H_{12}O_6 + 4O_2 \longrightarrow 3CO_2 + 6H_2O + 2CO + C$$

(ii) Carbon dioxide concentration.
(iii) Temperature. A high temperature will induce the rate of respiration while a low temperature will reduce the respiration rate.

13. Secondary metabolites do not induce the growth and development of plants. Their functions are more profound in that, they produce toxins that prevent herbivores, mammals and plant eating insects from damage.

14. Phytochrome is a blue protein pigment found in most etiolated plants. It's function is to detect and respond to the direction of light energy in etiolated plants.

15. (i) Provided the farmer grew pineapples that are eventually approaching maturity and have not yet started flowering, I would advise the farmer to spray the hormone, auxin, since it promotes flowering in pineapples.
(ii) Suppose the farmer has had a number of fruit and leaf droppings in his orchard, I would avise him to apply hormone auxin as it prevents leaf and fruit drop.